Smart Gate v3: Test Bench Design

1. Purpose

This document describes the 3D-printed test bench that holds the Smart Gate v3 hardware: why it is open and modular, where each part sits, how the sensors are mounted, its known limitations, and how to rebuild or improve it. CAD files are listed in Section 9 and kept in this folder (`docs/cad/`).

2. Design evolution

Version 1: enclosed case. The first design held every part in a compact case, with wiring hidden under a removable cover. That suits a finished product, because it protects the parts and looks tidy. For a test bench it caused problems: parts and wires were hidden, connections were hard to follow, and the fixed layout made it difficult to move or change anything.

Version 2: open modular bench (figure 1, current). The second design puts accessibility and modularity first (Figures 1–2). It has two perforated 3D-printed ABS decks resting on four corner pillars, each built from several printed sections joined together. Both decks share a grid of holes so parts can be fixed anywhere. The change was presented to OpenSI and approved at the sponsor meeting on 17 September 2026.

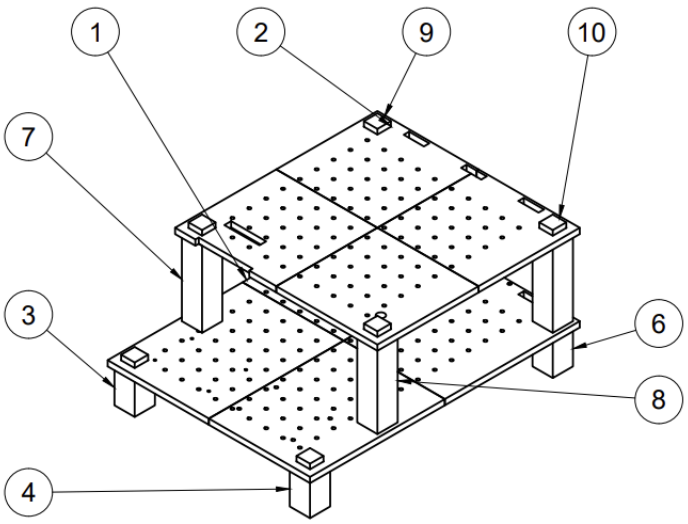


Figure 1: Final testbench design

> The inherited v0.2 enclosure drawing (Fusion 360, N. Drury, Oct 2024) is kept for reference at `docs/smartgate Drawing final v4.pdf`., see also figure 1 above

3. Component layout

Each part is placed by its function, how often it needs to be reached, and what it needs around it to work reliably.

Component	Location	Reason
Stepper motor and gate arm	Lower deck, front	Gate clearly visible, with room to move freely

Component	Location	Reason
Hall-effect sensors	Lower deck, on printed stands bolted through the hole grid	Detect the arm at the open and closed positions
Camera and PIR sensor	Upper deck, one side, in custom mounts	Clear, unobstructed view, away from cables and other hardware
Battery pack	Upper deck	Moved from the lower deck because cells need replacing and must be easy to reach
Jetson Orin Nano and L298N driver	Upper deck, centre	Close to the parts they connect to; wiring runs down through deck slots to the gate

4. Custom sensor mounts

Neither the camera nor the PIR sensor can stand upright on the bench by itself, so a mount was designed and printed for each (Figure 3).

- **Camera mount:** an upright plate with four raised standoffs that line up with the camera board's holes, so the board screws on securely. A slot gives the ribbon cable a clear path to the Jetson.
- **PIR mount:** an upright frame with a square opening for the lens and side guides that hold the board, keeping the sensor facing forward with nothing in front of it.

5. Design principles

- **Visibility:** every part sits in the open and can be identified at a glance.
- **Traceable wiring:** wires are visible and routed through deck slots to the gate below, so each connection can be followed from source to destination.
- **Modularity:** parts are attached with Velcro or bolted through the grid, so they can be moved or swapped freely.
- **Extensibility:** future teams can rearrange or replace any part without redesigning the bench.

6. Limitations and workarounds

Issue	Cause	Current workaround	Recommended fix
Fragile joints	Bench is printed in separate sections so it can be taken apart	Lower-deck legs hot-glued (removable if full disassembly is needed); upper deck left detachable for access	Thicker walls and joints in CAD, higher infill, PETG, heat-set threaded inserts
Hole grid does not match components	Bench and holes printed larger than intended; first full-size print	Velcro dots for all parts except the Hall stands	Redesign with hole spacing matched to each component; nuts under screws for a firmer fix
Loose wiring and exposed parts	Time and budget limits; open layout	Wired as tidily as possible	Longer wires routed along the bench, printed cable clips through the grid

7. Future improvements

- Matched hole pattern so parts screw down securely.
- Cable management with longer wires and printed clips.
- Colour coding of mounts by function (sensing, control, gate) so parts are identified at a glance.
- Overall finish: with these changes the bench would look more organised and be easier to present and to work on.

8. Decision record

The team chose to keep Version 2 rather than reprint, because reprinting would take time from integration work. Partial gluing fixed the fragility and Velcro fixed the size mismatch, so the bench is fully functional for testing. The improvements above are carried forward as Recommendation 8 in the Final Delivery Report.

9. CAD files

Designed in Fusion 360. STEP files open in any CAD tool for editing; STL files are ready to slice and print. Both are kept in `docs/cad/`.

Part	Qty	STEP file	STL file
Lower deck	1	`SGv3_lower_deck.step`	`SGv3_lower_deck.stl`
Upper deck	1	`SGv3_upper_deck.step`	`SGv3_upper_deck.stl`
Corner pillar section	4 pillars, several sections each	`SGv3_pillar_section.step`	`SGv3_pillar_section.stl`
Camera mount	1	`SGv3_camera_mount.step`	`SGv3_camera_mount.stl`
PIR mount	1	`SGv3_pir_mount.step`	`SGv3_pir_mount.stl`
Hall sensor stand	2	`SGv3_hall_stand.step`	`SGv3_hall_stand.stl`
Gate arm	1	`SGv3_gate_arm.step`	`SGv3_gate_arm.stl`

Print settings: ABS, 0.2 mm layer height, 20% infill, 3 walls, supports only where needed (mount standoffs and slots). Decks and pillars were split into sections so each fits the printer bed.

10. Figures

- Figure 1: bench, front view (`docs/cad/fig1_bench_front.jpg`)
- Figure 2: bench, side view (`docs/cad/fig2_bench_side.jpg`)
- Figure 3: camera and PIR mounts (`docs/cad/fig3_mounts.jpg`)